# A NOTE ON TWO CONJECTURES OF SUN

XIN WANG AND DASHENG WEI

Abstract. We prove the following two conjectures posed by Zhi-Wei Sun:

(i) If $a$ and $b$ are two coprime positive integers, then every positive rational number can be expressed as $ax^4 + by^4 + z^2 + w^2$ for some $x, y, z, w \in \mathbb{Q}$.

(ii) If $k$ is a totally real number field, then every positive number in $k$ can be expressed as $x^4 + y^4 + z^2 + w^2$ for some $x, y, z, w \in k$.

## 1. Introduction

Lagrange's four-square theorem, established in 1770, asserts that every positive integer can be expressed as a sum of four integer squares. An element in a number field $k$ is said to be *positive* if it is positive at all real places of $k$. In 1921, C.L. Siegel [Sie21] demonstrated that every positive integer in a totally real number field can be expressed as a sum of four integral squares. Siegel's result can easily be extended to a rational version as well, *i.e.*, every positive element in a totally real number field $k$ can be expressed as a sum of four squares in $k$. In [Sun22], Zhi-Wei Sun raised several conjectures which strengthen this rational version. The goal of this note is to prove two of Sun's conjectures.

**Theorem 1.1.** *Let $a$ and $b$ be non-zero coprime integers.*

*(i) If $a$ or $b$ is negative, then every $r \in \mathbb{Q}$ can be expressed as $ax^4 + by^4 + z^2 + w^2$ for some $x, y, z, w \in \mathbb{Q}$.*

*(ii) If both $a$ and $b$ are positive, then every positive number $r \in \mathbb{Q}$ can be expressed as $ax^4 + by^4 + z^2 + w^2$ for some $x, y, z, w \in \mathbb{Q}$.*

**Theorem 1.2.** *Let $k$ be a number field. Then every positive number $r \in k$ can be expressed as $x^4 + y^4 + z^2 + w^2$ for some $x, y, z, w \in k$.*

Theorem 1.1 (ii) is Conjecture 6.1 in [Sun22], and Theorem 1.2 confirms the second assertion in Conjecture 6.3 in [Sun22] (Assertion (i) of [Sun22, Conjecture 6.3] has been proved previously by Song and Xu [SX24]).

The positivity of $r$ ensures that it can be expressed in the required form at all archimedean places. If $k$ is a totally imaginary number field, every nonzero element in $k$ is understood as positive, we have the following corollary.

*Date*: May 23, 2025.

2010 *Mathematics Subject Classification.* 14G05 (11D57, 14F22).

**Corollary 1.3.** *Let $k$ be a totally imaginary number field. Then every $r \in k$ can be expressed as $x^4 + y^4 + z^2 + w^2$ for some $x, y, z, w \in k$.*

Our proofs of Theorems 1.1 and 1.2 will be based on the Hasse principle, with which we may reduce the problems to solving the relevant equations in the local completions of a number field. More precisely, let $k$ be a number field and let elements $a, b, c, d, r \in k^*$ be given. In Section 2, we use tools from arithmetic geometry to show that the solubility of the equation

$$ax^4 + by^4 + cz^2 + dw^2 = r$$

satisfies the Hasse principle. In fact, our Theorem 2.1 provides a slightly more general Hasse principle. Then, in Section 3 (resp. Section 4) we check the local solubility of the equation involved in Theorem 1.1 (resp. 1.2). To establish the desired Hasse principle, we observe that the equation under consideration defines a fibration of Châtelet surfaces. Colliot-Thélène, Sansuc and Swinnerton-Dyer [CTSSD87] showed that Châtelet surfaces satisfy weak approximation with Brauer–Manin obstruction by the descent theory developed in [CTS87]. Our proof of Theorem 2.1 builds upon this result for Châtelet surfaces and follows from the standard fibration method [Har94, HW16, HWW22].

**Acknowledgements.** We are grateful to F. Xu for insightful discussions which have been useful in particular for Lemma 4.2, and to the anonymous referees for their many valuable suggestions. The second author is supported by National Key R&D Program of China (Grant No. 2020YFA0712600) and National Natural Science Foundation of China (Grant Nos. 12371014 and 12231009).

## 2. A Hasse principle by the fibration method

Let $k$ be a number field and $\Omega_k$ the set of all places of $k$. For each place $v \in \Omega_k$, let $k_v$ denote the completion of $k$ at $v$. For an algebraic $k$-variety $X$, let $X(k)$ (resp. $X(k_v)$) denote the set of $k$-rational (resp. $k_v$-rational) points of $X$. We say that the Hasse principle holds for $X$ if the product $\prod_{v\in\Omega_k} X(k_v)$ being nonempty implies that $X(k)$ is nonempty. We say that weak approximation holds for X if X(k) is dense in the product space $\prod_{v\in\Omega_k} X(k_v)$. In our terminology, it is clear that weak approximation implies the Hasse principle. If X is smooth and $U \subset X$ is a dense open subset, then weak approximation holds for $X$ if and only if it holds for $U$. Let $\mathbf{A}_k$ be the adèle ring of $k$, and let $X(\mathbf{A}_k)$ denote the set of adelic points of an algebraic $k$-variety $X$ (cf. [CTS21, §13.1.3]). If $X(\mathbf{A}_k) \neq \varnothing$, then $\prod_{v\in\Omega_k} X(k_v) \neq \varnothing$. The converse is true if $X$ is a proper variety.

The Brauer group of $X$ is defined as the étale cohomology group $\mathrm{Br}(X) = H^2_{\text{ét}}(X, \mathbf{G}_m)$. Based on local class field theory, the Brauer-Manin pairing between $X(\mathbf{A}_k)$ and $\mathrm{Br}(X)$, which takes values in $\mathbb{Q}/\mathbb{Z}$, can be defined by using local

evaluation maps on Brauer classes. The Brauer–Manin set $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$ is the subset of $X(\mathbf{A}_k)$ consisting of the elements that are orthogonal to all elements of $\mathrm{Br}(X)$. Global class field theory implies that $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$ contains $X(k)$. Suppose that $X$ is a proper variety. We say that weak approximation with Brauer–Manin obstruction holds for $X$ if $X(k)$ is dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$. The reader is referred to [CTS21, §13.3] for more details on this subject.

Now let $X$ be a proper smooth $k$-variety equipped with a surjective morphism $f : X \to \mathbb{P}^1_k$ whose geometric generic fiber is rationally connected. Assume that weak approximation with Brauer–Manin obstruction holds for the smooth fibers of $f$ (see [HW16, HWW22, WXZ24] for a number of cases where this assumption on the smooth fibers holds). Then, in nice situations the so-called fibration method can be used to prove that $X$ satisfies weak approximation with Brauer–Manin obstruction. By this method, we obtain the following theorem.

**Theorem 2.1.** *Let $k$ be a number field and let $m \leq 4$ be a positive integer. Let $a, b, c, d, r \in k^*$ and let $U \subset \mathbb{A}^4_k$ be the affine variety defined by the equation*

$$ax^m + by^m + cz^2 + dw^2 = r.$$

*Let $X$ be a proper smooth variety containing $U$ as a dense open subset. Then weak approximation holds for $X$, and hence the Hasse principle holds for $U$.*

*Proof.* If $m \leq 2$, $X$ satisfies the Hasse principle and weak approximation by the Hasse-Minkowski theorem. So we may assume $m = 3$ or 4.

For any k-variety $Y$, we write $\overline{Y} = Y \times_k \overline{k}$, where $\overline{k}$ is an algebraic closure of $k$, and we denote its Picard group by $\mathrm{Pic}(\overline{Y})$. The algebraic Brauer group $\mathrm{Br}_1(Y)$ is the kernel of the natural map $\mathrm{Br}(Y) \to \mathrm{Br}(\overline{Y})$.

Let $U'$ be the open subset of $U$ defined by $cz^2 + dw^2 \neq 0$. Then $\mathrm{Pic}(\overline{U}') = 0$ since $\overline{U}'$ is isomorphic to an open subset of the product of $\mathbb{A}^2_k$ and $\mathbb{G}_m$. Let $\alpha = \sqrt{-d/c}$. Let $D_+$ and $D_-$ be respectively divisors of $U$ defined by $z + \alpha w = 0$ and $z - \alpha w = 0$. Notice that $U \setminus U' = \{D_+, D_-\}$. Using the inclusion $\overline{U}' \subset \overline{U}$, we have that $\mathrm{Pic}(\overline{U}) = 0$ and $\overline{k}[U]^\times = \overline{k}^*$. By Hochschild-Serre's spectral sequence, we have $\mathrm{Br}_1(U) = \mathrm{Br}_0(U)$. Since $\mathrm{Br}_1(X)$ is a subset of $\mathrm{Br}_1(U)$, we have $\mathrm{Br}_1(X) = \mathrm{Br}_0(X)$. Since $X$ is geometrically rational, we have $\mathrm{Br}(\overline{X}) = 0$. Therefore, $\mathrm{Br}(X) = \mathrm{Br}_0(X)$.

Let $V$ be the open subset of $U$ defined by $(x, y) \neq (0, 0)$. There is a surjective morphism $f : V \to \mathbb{P}^1_k$ defined by $(x, y) \to (x : y)$. The generic fiber of $f$ is an open subset of a Châtelet surface, so by [CTSSD87, Theorem B], the proper smooth compactification of each smooth fiber of $f$ satisfies weak approximation with Brauer-Manin obstruction. On the other hand, all geometric fibers of $f$ are integral. Therefore, by [Har94, Theorem 4.2.1] or [HWW22, Theorem 1.3], the proper smooth model $X$ of $V$ satisfies weak approximation with Brauer-Manin obstruction.

Since $\mathrm{Br}(X) = \mathrm{Br}_0(X)$, $X$ satisfies the Hasse principle and weak approximation. It follows that the open subset $U$ of $X$ satisfies the Hasse principle and weak approximation. □

## 3. Proof of Theorem 1.1

For any prime number $p$, let $\mathrm{ord}_p$ denote the normalized $p$-adic valuation on $\mathbb{Q}$ or $\mathbb{Q}_p$, and let $(a,b)_p$ denote the $p$-adic 2nd Hilbert symbol for elements $a, b \in \mathbb{Q}_p^*$.

**Lemma 3.1.** *Let $p$ be a prime number and let $a, b, r \in \mathbb{Q}_p^*$. If $p, \mathrm{ord}_p(a)$ and $\mathrm{ord}_p(b)$ are all odd, we assume that $p \equiv 1 \mod 4$. Then the equation*

$$ax^4 + by^4 + z^2 + w^2 = r \tag{3.1}$$

*is solvable in $\mathbb{Q}_p$.*

*Proof.* Suppose that $p$ is odd. By our assumption, either $p \equiv 1 \mod 4$, or $\mathrm{ord}_p(a)$ is even (or $\mathrm{ord}_p(b)$ is even and we may give a similar proof). We always have $(-1,-a)_p = 1$. We may choose $y = 0$ and choose $x_0 \in \mathbb{Q}_p^*$ such that $\mathrm{ord}_p(x_0^4) < \mathrm{ord}_p(r) - \mathrm{ord}_p(a)$. So

$$(-1, r - ax_0^4)_p = (-1,-a)_p = 1,$$

hence $ax_0^4 + z^2 + w^2 = r$ is solvable in $\mathbb{Q}_p$, so (3.1) is solvable in $\mathbb{Q}_p$.

In the following we assume $p = 2$. If $(-1,-a)_2 = 1$ or $(-1,-b)_2 = 1$, then the result can be proved similarly as in the previous case.

Now we assume that $(-1,-a)_2 = (-1,-b)_2 = -1$, *i.e.*, $(-1,a)_2 = (-1,b)_2 = 1$. Since $b$ is a sum of two squares, $b(z^2+w^2)$ is also a sum of two squares, it is sufficient to show that $a/bx^4 + y^4 + z^2 + w^2 = r/b$ is solvable in $\mathbb{Q}_2$. Therefore, we only need to show

$$ax^4 + y^4 + z^2 + w^2 = r \tag{3.2}$$

is solvable in $\mathbb{Q}_2$ for any $a \in \mathbb{Q}_p^*$ such that $(-1,a)_2 = 1$ and $r \in \mathbb{Z}_2 \setminus \{0\}$. We only need to verify this claim for $r = 2^i r'$ and $a = 2^j a'$ with $0 \le i, j \le 3$, $a', r' \in \mathbb{Z}_2$ such that $2 \nmid r'a'$ and $a' \equiv 1 \mod 4$. If $r' \equiv 1 \mod 4$, then since $(-1,r)_2 = 1$, $r$ is a sum of two squares in $\mathbb{Q}_2$ and hence (3.2) is solvable in $\mathbb{Q}_2$. Therefore, we may assume $r' \equiv 3 \mod 4$.

First we consider the cases that $i$ or $j$ is odd.

(i) Suppose $i = 1$. Since $r - 1 \equiv 5 \mod 8$, we have $(-1, r-1)_2 = 1$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x,y) = (0,1)$.

(ii) Suppose $i = 3$. Note that $r - 16 = 2^3(r' - 2)$, $r' - 2 \equiv 1 \mod 4$, hence $(-1, r-16)_2 = 1$, so (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x,y) = (0,2)$.

(iii) Suppose $j = 1$. Then $a + 1 \equiv 3 \mod 8$, so $(-1, -(a+1))_2 = 1$. Since $(a + 1 - 16r)/(a + 1)$ is a square in $\mathbb{Z}_2$, we have

$$(-1, -(a+1)/16 + r)_2 = (-1, -(a+1))_2 = 1,$$

hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1/2, 1/2)$.

(iv) Suppose $j = 3$. Let $x_0 = 1/4$ and $y_0 = 1/2$. Then $r - ax_0^4 - y_0^4 = r - (a'+2)/32$. Note that $-(a' + 2) \equiv 1 \mod 4$, hence $(-1, r - (a' + 2)/32)_2 = 1$, and this implies that (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1/4, 1/2)$.

Now the remaining case is that both $i$ and $j$ are even.

(v) Suppose $i = j = 0$. Then $r - a - 1 \equiv 1 \mod 4$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1, 1)$.

(vi) Suppose $i = 2$ and $j = 0$. If $a' \equiv 5 \mod 8$, then $-(a + 1) \equiv 2 \mod 8$. Thus $(-1, r-(a+1)/16)_2 = 1$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1/2, 1/2)$. If $a' \equiv 1 \mod 8$, then $r - a - 1 \equiv 2 \mod 8$. Thus $(-1, r - a - 1)_2 = 1$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1, 1)$.

(vii) Suppose $i = 0$ and $j = 2$. If $r \equiv 3 \mod 8$, then $r - 1 \equiv 2 \mod 8$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (0, 1)$. If $r \equiv 7 \mod 8$, then $r - a - 1 \equiv 2 \mod 8$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1, 1)$.

(viii) Suppose $i = j = 2$. If $r' \equiv -a' (\equiv 3 \text{ or } 7) \mod 8$, then

$$r - a - 16 \equiv 8(r' - 2) \equiv 8 \mod 32,$$

hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1, 2)$. If $r' \not\equiv -a' \mod 8$, then $r - a \equiv 8 \mod 32$, hence (3.2) is solvable in $\mathbb{Q}_2$ by choosing $(x, y) = (1, 0)$. □

The proof of Theorem 1.1 follows Theorem 3.2 below.

**Theorem 3.2.** *Let $a$ and $b$ be non-zero rational numbers. For any odd prime $p$, if both $\mathrm{ord}_p(a)$ and $\mathrm{ord}_p(b)$ are odd, we assume that $p \equiv 1 \mod 4$.*

*(i) If $a$ or $b$ is negative, then every $r \in \mathbb{Q}$ can be expressed as $ax^4 + by^4 + z^2 + w^2$ for some $x, y, z, w \in \mathbb{Q}$.*

*(ii) If both $a$ and $b$ are positive, then every $r \in \mathbb{Q}_{\geq 0}$ can be expressed as $ax^4 + by^4 + z^2 + w^2$ for some $x, y, z, w \in \mathbb{Q}$.*

*Proof.* We may assume $r \neq 0$. Under the assumptions of $a$, $b$ and $r$, (3.1) is solvable in $\mathbb{R}$. By Lemma 3.1, (3.1) is locally solvable at any finite place. Therefore, by Theorem 2.1, (3.1) is solvable in $\mathbb{Q}$. □

**Remark 3.3.** If there exists a prime $p \equiv 3 \mod 4$ such that both $\mathrm{ord}_p(a)$ and $\mathrm{ord}_p(b)$ are odd, then possibly some positive rational number $r$ cannot be expressed as $ax^4 + by^4 + z^2 + w^2$. For example, 6 cannot be expressed as $3x^4 + 27y^4 + z^2 + w^2$ in $\mathbb{Q}$ since it cannot be in $\mathbb{Q}_3$.

Many positive integers $r$ cannot expressed as the form $x^4+y^4+z^2+w^2$ with $x,y,z,w\in\mathbb{Z}$, *e.g.*, $r=23,44,\cdots$. Therefore, we cannot expect an integral version of Theorem 1.1 (ii) and 1.2. However, we have the following conjecture which has been verified by computer for $|r|\leq 10^6$.

**Conjecture 3.4.** *Every integer $r$ can be expressed as $x^3+y^3+z^2+w^2$ for some $x,y,z,w\in\mathbb{Z}$.*

## 4. Proof of Theorem 1.2

Let $k$ be a number field, and let $v$ be a finite place of $k$. Let $\mathfrak{o}_v$ be the valuation ring of $k_v$ with the maximal ideal $\mathfrak{p}$. Let $\epsilon\in\mathfrak{o}_v^*$ be a unit. The defect $\boldsymbol{\delta}(\epsilon)$ of $\epsilon$ is the smallest idea $\mathfrak{p}^d$, with $d\in\mathbb{N}\cup\{+\infty\}$, such that $\epsilon\equiv u^2 \mod \mathfrak{p}^d$ for some $u\in\mathfrak{o}_v^*$ (see [O'M63, §63A]).

**Proposition 4.1.** [O'M63, 63:2] *Let $\epsilon$ be a unit in $\mathfrak{o}_v$. If $k_v$ is non-dyadic, then $\boldsymbol{\delta}(\epsilon)$ is $0$ or $\mathfrak{o}_v$; if $k_v$ is dyadic, then $\boldsymbol{\delta}(\epsilon)$ is one of the ideals*

$$0\subset 4\mathfrak{o}_v\subset 4\mathfrak{p}^{-1}\subset 4\mathfrak{p}^{-3}\subset\cdots\subset\mathfrak{p}.$$

Let $\mathrm{ord}_v$ denote the normalized discrete valuation on $k_v$, let $(\cdot,\cdot)_v$ denote the 2nd Hilbert Symbol on $k_v$. Let $e=\mathrm{ord}_v(2)$. The following lemma is in fact [Xu93, Remark 1], here we give a simple proof similar to that of [Hsi75, Lemma 4].

**Lemma 4.2.** *Let $\epsilon$ and $\eta$ be units in $\mathfrak{o}_v$ with defects $\boldsymbol{\delta}(\epsilon)=\mathfrak{p}^s$ and $\boldsymbol{\delta}(\eta)=\mathfrak{p}^t$. If $s+t>2e$, then*

$$(\epsilon,\eta)_v=1.$$

*Proof.* We only need to show that the equation $\epsilon x^2+\eta y^2=1$ is solvable in $k_v$. We may write $\epsilon=s_1^2+\delta_1$ and $\epsilon=s_2^2+\delta_2$ with each $s_i\in\mathfrak{o}_v^*$, $\delta_1\in\mathfrak{p}^s$ and $\delta_2\in\mathfrak{p}^t$. Let $\delta_i'=\delta_i/s_i$ for $i=1,2$, it suffices that $(1+\delta_1')x^2+(1+\delta_2')y^2=1$ is solvable in $k_v$. If $\delta_1'$ or $\delta_2'$ is equal to 0, obviously $(1+\delta_1')x^2+(1+\delta_2')y^2=1$ is solvable in $k_v$. So we may assume that $\delta_1'\delta_2'\neq 0$. Note that $s+t>2e$, the equation $\delta_1'w^2-2w+\delta_2'=0$ has a solution $\alpha\neq -1$ in $k_v$ since its discriminant is nonzero and a square. Let $x=\alpha y$, hence

$$(1+\delta_1')x^2+(1+\delta_2')y^2=(\alpha^2+1+\delta_1'\alpha^2+\delta_2')y^2=(\alpha^2+1+2\alpha)y^2=(1+\alpha)^2y^2.$$

Let $y=(1+\alpha)^{-1}$, then we get a solution of $(1+\delta_1')x^2+(1+\delta_2')y^2=1$. □

**Lemma 4.3.** *For any $r\in k_v^*$, the equation*

$$x^4+y^4+z^2+w^2=r \tag{4.1}$$

*is solvable in $k_v$.*

*Proof.* Suppose $v \nmid 2$. Then $(-1,-1)_v = 1$. By choosing $\mathrm{ord}_v(x) < \mathrm{ord}_v(r)/4$ and $y = 0$, we see that (4.1) is solvable in $k_v$.

Suppose $v \mid 2$. Let $e = \mathrm{ord}_v(2)$. If $k_v/\mathbb{Q}_2$ has even degree, then $(-1,-1)_v = 1$, and similarly as above (4.1) is solvable in $k_v$. Now we assume that $[k_v : \mathbb{Q}_2]$ is odd, then $e$ is odd and $(-1,-1)_v = -1$. We only need to show that (4.1) is solvable in $k_v$ for any $\bar{r} \in k_v^*/k_v^{*4}$. The valuation of $2^i$ is $ei$. Note that $e$ is odd, for any element in $k_v^*/k_v^{*4}$, we may choose its representative to be of the form $r = 2^i r'$ with $0 \le i \le 3$ and $r' \in \mathfrak{o}_v^*$. Hence we only need to show that (4.1) is solvable in $k_v$ for any such $r = 2^i r'$. If $(-1, r')_v = 1$, then (4.1) is solvable in $k_v$ (let $(x, y) = (0, 0)$). So, we may always assume that $(-1, r')_v = -1$. We may write $r' = s^4(1 + \pi')$ with $\mathrm{ord}_v(\pi') \ge 1$, hence $(-1, 1 + \pi')_v = -1$.

(i) Suppose $i = 0$, *i.e.*, $r = r'$. Then $r - 2s^4 = -s^4(1 - \pi')$. We have

$$(1 + \pi')(1 - \pi') = 1 - \pi'^2 = 1 + \pi'^2 - 2\pi'^2 = (1 + \pi')^2 - 2\pi' - 2\pi'^2,$$

so $\boldsymbol{\delta}((1+\pi')(1-\pi')) \subset \mathfrak{p}^{1+e}$. On the other hand, $\boldsymbol{\delta}(-1) = \mathfrak{p}^e$. Note that $1+e+e > 2e$, so by Lemma 4.2, we have

$$(-1, (1 + \pi')(1 - \pi'))_v = 1,$$

hence

$$(-1, 1 - \pi')_v = (-1, 1 + \pi')_v = -1.$$

Therefore,

$$(-1, r - 2s^4)_v = (-1,-1)_v(-1, 1 - \pi')_v = 1,$$

which implies (4.1) is solvable in $k_v$ by choosing $x = y = s$.

(ii) Suppose $i = 1$, *i.e.*, $r = 2s^4(1 + \pi')$. Then $r - s^4 = s^4(1 + 2\pi')$, hence $\boldsymbol{\delta}(r - s^4) \subset \mathfrak{p}^{1+e}$, so by Lemma 4.2, we have $(-1, r - s^4)_v = 1$. Therefore, (4.1) is solvable in $k_v$ by choosing $(x, y) = (s, 0)$.

(iii) Suppose $i = 2$, *i.e.*, $r = 4s^4(1 + \pi')$. Then $r - 2s^4 = 2s^4(1 + 2\pi')$, hence $\boldsymbol{\delta}(1 + 2\pi') \subset \mathfrak{p}^{1+e}$, so by Lemma 4.2, we have $(-1, 1 + 2\pi')_v = 1$, hence $(-1, r - 2s^4)_v = 1$. Therefore, (4.1) is solvable in $k_v$ by choosing $x = y = s$.

(iv) Suppose $i = 3$, *i.e.*, $r = 8s^4(1 + \pi')$. Then $r - 16s^4 = -8s^4(1 - \pi')$. By the discussion as in case (i), we have $(-1, 1 - \pi')_v = (-1, 1 + \pi')_v = -1$. Hence $(-1, r - 16s^4) = 1$, so (4.1) is solvable in $k_v$ by choosing $(x, y) = (2s, 0)$. □

*Proof of Theorem 1.2.* By the positivity assumption of $r$, (4.1) is solvable at any archimedean place. By Lemma 4.3, (4.1) is locally solvable at any finite place. By Theorem 2.1, (4.1) is solvable in $k$. □

## References

[CTS87] J.-L. Colliot-Thélène and J.-J. Sansuc. La descente sur les variétés rationnelles. II. *Duke Math. J.*, 54(2):375–492, 1987.

[CTS21] J.-L. Colliot-Thélène and A. N. Skorobogatov. *The Brauer-Grothendieck group*, volume 71 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge.* Springer, Cham, 2021.

[CTSSD87] J.-L. Colliot-Thélène, J.-J. Sansuc, and P. Swinnerton-Dyer. Intersections of two quadrics and Châtelet surfaces. I,II. *J. Reine Angew. Math.*, 373:37–107; ibid. 374: 72–168, 1987.

[Har94] D. Harari. Méthode des fibrations et obstruction de Manin. *Duke Math. J.*, 75(1):221–260, 1994.

[Hsi75] J. S. Hsia. Spinor norms of local integral rotations. I. *Pacific J. Math.*, 57(1):199–206, 1975.

[HW16] Y. Harpaz and O. Wittenberg. On the fibration method for zero-cycles and rational points. *Ann. of Math. (2)*, 183(1):229–295, 2016.

[HWW22] Y. Harpaz, D. Wei, and O. Wittenberg. Rational points on fibrations with few non-split fibres. *J. Reine Angew. Math.*, 791:89–133, 2022.

[O'M63] O. T. O'Meara. *Introduction to quadratic forms*, volume Band 117 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin-Göttingen-Heidelberg; Academic Press, Inc., Publishers, New York, 1963.

[Sie21] C. L. Siegel. Darstellung total positiver Zahlen durch Quadrate. *Math. Z.*, 11:246–275, 1921.

[Sun22] Z. Sun. Sums of four rational squares with certain restrictions. *arXiv:2010.05775v7*, 2022.

[SX24] H. Song and F. Xu. Proof of one of Sun Zhi-Wei's conjectures. *Scientia Sinica Mathematica*, 54(9):1345–1356, 2024.

[Xu93] F. Xu. Integral spinor norms in dyadic local fields. I. *Pacific J. Math.*, 157(1):179–200, 1993.

School of Mathematics and Statistics, Beijing Technology and Business University , Beijing 102488, P. R. China

*Email address*: wangxin@btbu.edu.cn

Academy of Mathematics and System Science, CAS, Beijing 100190, P. R. China *and* School of mathematical Sciences, University of CAS, Beijing 100049, P. R. China

*Email address*: dshwei@amss.ac.cn